

Experiment 4

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Slot: B

Title: Exploration vs. Exploitation Analysis using Multi-Armed Bandit Algorithms

Software Required:

- Python
- NumPy
- Matplotlib

AIM

To implement the Multi-Armed Bandit algorithm using the ϵ -Greedy (Epsilon-Greedy) method and analyze the trade-off between exploration and exploitation.

THEORY

- The Multi-Armed Bandit problem is a Reinforcement Learning technique where an agent must choose between multiple actions (arms) to maximize the total reward.
- **Exploration:** Trying different actions to gain more information.
- **Exploitation:** Choosing the action that currently gives the highest reward.
- The ϵ -Greedy algorithm balances these two strategies.

PROCEDURE

1. Open Python IDLE.
2. Import NumPy and Matplotlib.
3. Define the reward probabilities for each arm.
4. Set the epsilon (ϵ) value.
5. Initialize action counts and estimated rewards.
6. Run the ϵ -Greedy algorithm for several iterations.
7. Update the estimated rewards after each action.
8. Display the number of times each arm is selected.
9. Plot the results using Matplotlib.

10. Analyze the exploration and exploitation behavior.

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)

# Probability of reward for each arm
true_rewards = [0.2, 0.5, 0.75, 0.9]

n_arms = len(true_rewards)
epsilon = 0.1
iterations = 1000

Q = np.zeros(n_arms)
N = np.zeros(n_arms)

total_reward = 0
rewards = []

for i in range(iterations):

    if np.random.rand() < epsilon:
        action = np.random.randint(n_arms)      # Exploration
    else:
        action = np.argmax(Q)                  # Exploitation

    reward = 1 if np.random.rand() < true_rewards[action] else 0

    N[action] += 1
    Q[action] += (reward - Q[action]) / N[action]

    total_reward += reward
    rewards.append(total_reward)

print("Estimated Reward of Each Arm")

for i in range(n_arms):
    print("Arm", i+1, ":", round(Q[i],3),
          " Selected", int(N[i]), "times")
```

```
print("\nTotal Reward =", total_reward)
plt.figure(figsize=(7,5))
plt.plot(rewards)
plt.title("Exploration vs Exploitation using  $\epsilon$ -Greedy")
plt.xlabel("Iterations")
plt.ylabel("Cumulative Reward")
plt.grid(True)
plt.show()
```

OUTPUT:

Estimated Reward of Each Arm

Arm 1 : 0.2 Selected 30 times

Arm 2 : 0.367 Selected 30 times

Arm 3 : 0.742 Selected 31 times

Arm 4 : 0.902 Selected 909 times

Total Reward = 860

